

Appendix A

Osteoporosis Dataset Description

Osteoporosis is a health condition that weakens bones, making them fragile and more likely to break.

The dataset is provided in file “Osteoporosis Dataset 600 samples.xlsx”.

Total number of dataset samples: 600 records of patients.

Classes (labels)

VBX: 0 – Healthy (501 samples); 1- Fracture (99 samples).

Features (25)

<i>Sex;</i>	<i>Age;</i>	<i>Ethnic;</i>	<i>Weight;</i>	<i>Height;</i>	<i>BMI</i>	<i>Waist;</i>
<i>Waist_category;</i>	<i>Menopause;</i>	<i>HRT;</i>	<i>Smoking;</i>	<i>ReumatoidArthritis;</i>		
<i>SecondaryOsteoporsis;</i>	<i>Alcohol;</i>	<i>Alcohol24;</i>	<i>VitaminD;</i>	<i>Calcium;</i>		
<i>Dose_walk;</i>	<i>Dose_moderate;</i>	<i>Dose_vigorous;</i>	<i>Dose_pleasure;</i>			
<i>Dose_sport;</i>	<i>Dose_execise;</i>	<i>Dose_lightDIY;</i>	<i>Dose_heavyDIY</i>			

Abbreviations

VBX – Vertebrate Fracture;

BMI - Body Mass Index;

HRT - Heart Rate;

DIY – ‘Do It Yorself’ activity

Blanks denote missing values.

When you deal with the missingness in the Pre-processing stage, if the missing values of a feature are > 45%, you are allowed to exclude it from consideration.